Supporting Information

Bayesian total evidence dating reveals the recent crown radiation of penguins

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1 Methods

1.1 Prior distributions for parameters

We used the following prior distributions for the parameters in all analyses except for the sensitivity analyses.

For the net diversification rate, d, we place a log-normal prior distribution with parameters[1] $M = -3.5$ and $S^2 = 1.5$. The 2.5% and 97.5% quantiles of this distribution are $1.6 \cdot 10^{-3}$ and 0.57 implying that it well covers the interval (with the lowest 5% quantile of 0.02 and the largest 95% quantile of 0.15) estimated in [1] Figure 1.

We place a Uniform(0,1) prior distribution for the turnover, ν, and Uniform(0,1) prior distribution for the fossil sampling proportion, s.

For birth, death and sampling rates (λ, μ, and ψ) we used log-normal distributions with parameters[1] $M = -2.5$ and $S^2 = 1.5$ and 2.5% and 97.5% quantiles of $4.34 \cdot 10^{-3}$ and 1.55.

For the time of origin, t_{or}, we use the Uniform prior distribution between the oldest sample and 160 Ma because Jetz *et al.* estimated the upper bound for the MRCA of all birds to be

[1]Parameters M and S for log-normal distributions here are the mean and standard deviation of the associated normal distributions

150 Ma ([1] Figure 1) and Lee *et al.* [2] estimated the origin of the Avialae clade to about 160 Ma (uncertainty is not reported) in the analysis of morphological data of dinosaurs.

We used a broad lognormal distribution where the 95% probability interval spanned $[4 \cdot 10^{-4}, 0.07]$ which is estimated for extinct dinosaurs in units of changes per character per Myr [2] for the rate of morphological evolution, μ_{morph} (the mean of the log-normal distribution of the morphological clock rates). The parameters[1] of this log-normal distribution are $M = -5.5$ and $S^2 = 2$, 2.5% and 97.5% quantiles are $8.11 \cdot 10^{-5}$ and 0.21, respectively.

μ_{mol} — the rate of molecular evolution was assigned a lognormal prior with parameters[1] $M = -3.5$ and $S^2 = 1$ with 5% and 95% quantiles of $4.25 \cdot 10^{-3}$ and 0.21.

We placed a Gamma distribution with parameters $\alpha = 0.5396$ and $\beta = 0.3819$ for the standard deviation of the uncorrelated lognormal relaxed models for both molecular and morphological data.

The gamma shape parameter on rate-variation across sites was assigned a Uniform[0,10] prior (for both molecular and morphological data).

1.2 Data

The number of characters with different numbers of states is given below:

State #	2	3	4	5	6	7
Character #	130	50	13	4	4	1

Out of 130 characters with two character states 48 have description 'present or absent'. The number of sites in each region is as follows:

Gene	RAG-1	12S	16S	COI	cytochrome b
Site #	2682	1143	1551	1105	1664

1.3 Fossil ages

We used the following age ranges for fossil penguins in our analyses (a compact summary is also given in Table 1).

Taxon: *Anthropornis grandis.*
Age: 34–52.5 Ma. Specimens are known from Telm 4 and Telm 7 of the La Meseta Formation [3]. Strontium dates from the stratigraphic column of Reguero et al. [4] provide ages for the bottom of Telm 4 and the top of Telm 7, which are applied as bounds.

Taxon: *Anthropornis nordenskjoeldi.*
Age: 34–52.5 Ma. The oldest specimens assigned to this species are potentially from Telm 4

or Telm 6 of the La Meseta Formation, while the specimens certainly referable to this species are known from Telm 7 [3]. Strontium dates from the stratigraphic column of Reguero et al. [4] provide ages for the bottom of Telm 4 and the top of Telm 7, which are applied as bounds.

Taxon: *Archaeospheniscus lopdelli.*
Age: 26–30Ma. The holotype and only described specimen comes from the Kokoamu Greensand, which is dated to between 26 Ma and 30 Ma (see [5], Figure 1).

Taxon: *Archaeospheniscus lowei.*
Age: 26–30 Ma. All three described specimens come from the Kokoamu Greensand, which is dated to between 26 Ma and 30 Ma (see [5], Figure 1).

Taxon: Burnside *"Palaeeudyptes"* (this name is a placeholder for a species that has not yet been formally named).
Age: 36–38.4 Ma. This fossil is from a section of the Burnside Formation that can be assigned to the Kaiatan NZ local stage (see [5], Figure 1).

Taxon: *Delphinornis arctowskii.*
Age: 34–41 Ma. We considered only tarsometatarsi to be firmly referable to this species. Such specimens are known only from Telm 7 of the La Meseta Formation [3]. Using the stratigraphic column of Reguero et al. [4], strontium dates for the top and bottom of Telm 7 are applied as bounds.

Taxon: *Delphinornis gracilis.*
Age: 34–41 Ma. We considered only tarsometatarsi to be firmly referable to this species. Such specimens are known only from Telm 7 of the La Meseta Formation [3]. Using the stratigraphic column of Reguero et al. [4], strontium dates for the top and bottom of Telm 7 are applied as bounds.

Taxon: *Delphinornis larseni.*
Age: 34–52.5 Ma (species range).
We considered only tarsometatarsi to be firmly referable to this species. Such specimens are known from Telm 5 and Telm 7 of the La Meseta Formation [3]. Strontium dates for the top of Telm 7 and (because there was no strontium date for the base of Telm 5) the base of Telm 4 (which is very short and certainly < 1 Ma) are used as bounds.

Taxon: *Delphinornis wimani.*
Age: 34–52.5 Ma. We considered only tarsometatarsi to be firmly referable to this species. Such specimens are known from Telm 5 and Telm 7 of the La Meseta Formation [3]. Strontium dates for the top of Telm 7 and (because there was no strontium date for the base of Telm 5) the base of Telm 4 (which is very short and certainly < 1 Ma) are used as bounds.

Taxon: *Duntroonornis parvus.*
Age: 21.7–30 Ma. The holotype comes from the Kokoamu Greensand and some referred

material comes from the Otekaike Limestone. Ages for the base of the Kokoamu Greensand and top of the Otekaike Limestone are used as bounds (see [5], Figure 1).

Taxon: *Eretiscus tonnii.*
Age: 16–21 Ma. A possible age range of 16–21 Ma is given based on Figure 2 of Cione et al. [6].

Taxon: *Icadyptes salasi.*
Age: 35.7–37.2 Ma. This age is based on two radiometrically dated layers that are believed to be correlated with layers above and below the holotype and only reported specimen [7].

Taxon: *Inkayacu paracasensis.*
Age: 35.7–37.2 Ma. This age is based on two radiometrically dated layers that are believed to be correlated with layers above and below the holotype and only reported specimen [7].

Taxon: *Kairuku grebneffi.*
Age: 26–30 Ma. The holotype and one referred specimen come from the Kokoamu Greensand, which is dated to between 26 Ma and 30 Ma (see [5], Figure 1).

Taxon: *Kairuku waitaki.*
Age: 26–30 Ma. The holotype and only described specimen comes from the Kokoamu Greensand, which is dated to between 26 Ma and 30 Ma (see [5], Figure 1).

Taxon: *Madrynornis mirandus.*
Age: 9.70–10.3 Ma. The holotype and only known specimen is believed to be 10 ± 0.3 Ma in age based on strontium dates from the "Entrerriens" sequence of the Puerto Madryn Formation where the fossil was collected (see [8]).

Taxon: *Marambiornis exilis.*
Age: 34–41 Ma. We considered only tarsometatarsi to be firmly referable to this species. Such specimens are known only from Telm 7 of the La Meseta Formation [3]. Using the stratigraphic column of Reguero et al. [4], strontium dates for the top and bottom of Telm 7 are applied as bounds.

Taxon: *Marplesornis novaezealandiae.*
Age: 5.33–15.97 Ma. This is one of the most difficult penguin fossils to date, because the specimen was found in a loose boulder on a beach that eroded out of a cliff of Great Sandstone deposits. Best estimates for the range of the Greta Siltstone from which the boulder is derived is middle-late Miocene [9], so a range of 5.33–15.97 is used.

Taxon: *Mesetaornis polaris.*
Age: 34–41 Ma. We considered only tarsometatarsi to be firmly referable to this species. Such specimens are known only from Telm 7 of the La Meseta Formation [3]. Using the stratigraphic column of Reguero et al. [4], strontium dates for the top and bottom of Telm 7 are applied as

bounds.

Taxon: *Pachydyptes ponderosus.*
Age: 34.5–36 Ma. All known fossils of this species are all assigned to the Runangan NZ local stage, the bounds of which are used for the age range (see [5], Figure 1).

Taxon: *Palaeeudyptes antarcticus.*
Age: 30.1–34.5. The holotype and only firmly referred specimen comes from the upper Ototara Limestone, which is assigned, to the Subbotina angiporoides zone of the local Whaingaroan Stage, which is in turn dated to between 30.1–34.5 (see [5], Figure 1).

Taxon: *Palaeeudyptes gunnari.*
Age: 34–54 Ma. Specimens are known from Telm 3 and Telm 7 (top) of the La Meseta Formation [3]. Strontium dates for the top of Telm 7 and (because there was no strontium date for the base of Telm 3) the base of Telm 2 (which is very short and certainly < 1 Ma) are used as bounds.

Taxon: *Palaeeudyptes klekowski.*
Age: 34–52.5 Ma (species range). We included the oldest potentially referable specimen from Telm 5 of the La Meseta Formation as well as the youngest specimens from Telm 7 [3] in the age range. Strontium dates for the top of Telm 7 and (because there was no strontium date for the base of Telm 5) the base of Telm 4 (which is very short and certainly < 1 Ma) are used as bounds.

Taxon: *Palaeospheniscus bergi.*
Age: 9.7–21 Ma. Specimens are known from the Gaiman and Chenque Formations, which are contemporaneous, as well as a young record from the Puerto Madryn Formation [10]. The age range extends from 21 Ma for the base of Gaiman Formation (Figure 2 of Cione et al. [6]) to 9.7 Ma for the Puerto Madryn Formation record [8].

Taxon: *Palaeospheniscus biloculata.*
Age: 16–21 Ma. A possible age range of 16–21 Ma is given based on Figure 2 of Cione et al. [6] representing the duration of the Gaiman Formation.

Taxon: *Palaeospheniscus patagonicus.*
Age: 16–21 Ma. A possible age range of 16–21 Ma is given based on Figure 2 of Cione et al. [6] representing the duration of the Gaiman Formation.

Taxon: *Paraptenodytes antarcticus.*
Age: 21–23 Ma. All known specimens are from the Monte Leon Formation. Chavez [11] showed that reports of this species from the Bahia Inglesia were erroneous.

Taxon: *Perudyptes devriesi.*
Age: 38–46 Ma. An age of $\sim$ 42 Ma for the holotype and only known specimen is based

on stratigraphic correlations and biostratigraphy [7, 12]. Given the lack of directly datable layers at the type locality, we extend this range by 4 Ma in either direction to accommodate uncertainty.

Taxon: *Platydyptes novazealandiae.*
Age: 23–26 Ma. All known specimens come from Oligocene sections of the Otekaike Limestone (see [13]).

Taxon: *Platydyptes marplesi.*
Age: 23–30 Ma. Specimens are known from the Kokoamu Greensand, providing the lower bound and the Otekaike Limestone, providing the upper bound of the age range (see [13]).

Taxon: *Pygoscelis grandis.*
Age: 2.5–8.6 Ma. The oldest specimens are from the Bonebed Member of the Bahia Inglesia Fm. and are from below a layer dated to 7.6 ± 1.3 Ma, whereas younger specimens are estimated to be 2.5–4.6 Ma based on biostratigraphy [14].

Taxon: *Spheniscus megaramphus.*
Age: 6.3–10 Ma. Stucchi [15] reported this specimens of this species in the Montemar Norte locality and Chavez [11] reported material from the Bonebed Member of the Bahia Inglesia Fm. Montemar Norte was estimated to be 10 Ma by Stucchi [16], providing an upper age limit. Bonebed Member specimens are near a radiometric layer dated to 7.6 ± 1.3 Ma, providing the lower bound.

Taxon: *Spheniscus muizoni.*
Age: 9–9.2 Ma. The sole specimen is dated to between 9 and 9.2 Ma based on radiometric dates above and below the specimen [17].

Taxon: *Spheniscus urbinai.*
Age: 5.7–9.63 Ma. Reported specimens have been recovered from the Sacaco, Sacaco Sur, Montemar, Aguada de Lomas and El Jajuay sites in the Pisco Formation acceding to Stucchi [15]. The 5.7 Ma radiometric date for the top of Sacaco is based on [18] and the 9.38 Ma radiometric date for El Jahuay is taken from [17].

Taxon: *Waimanu manneringi.*
Age: 60.5–61.6 Ma. The sole specimen is constrained to between 60.5Ma and 61.6Ma based on biostratigraphy [19].

Taxon: *Waimanu tuatahi.*
Age: 56–60.5 Ma. The most well-constrained fossil of this taxon is dated to 58–60 Ma based on biostratigraphy [19]. Other referred fossils are less well-constrained and a conservative range would be 56–60.5 is specified with an upper age limit based on the age of *Waimanu manneringi* which was collected lower in the stratigraphic column and a lower age limit set to the Paleocene-Eocene boundary, as the fossils are all of Paleocene age.

Table 1: The stratigraphic age ranges used for phylogenetic analyses of the penguin dataset.

#	Fossil taxon	Age interval
1	*Anthropornis grandis*	[34, 52.5]
2	*Anthropornis nordenskjoeldi*	[34, 52.5]
3	*Archaeospheniscus lopdelli*	[26, 30]
4	*Archaeospheniscus lowei*	[26, 30]
5	Burnside *"Palaeudyptes"*	[36, 38.4]
6	*Delphinornis arctowskii*	[34, 41]
7	*Delphinornis gracilis*	[34, 41]
8	*Delphinornis larseni*	[34, 52.5]
9	*Delphinornis wimani*	[34, 52.5]
10	*Duntroonornis parvus*	[21.7, 30]
11	*Eretiscus tonnii*	[16, 21]
12	*Icadyptes salasi*	[35.7, 37.2]
13	*Inkayacu paracasensis*	[35.7, 37.2]
14	*Kairuku grebneffi*	[26, 30]
15	*Kairuku waitaki*	[26, 30]
16	*Madrynornis mirandus*	[9.7, 10.3]
17	*Marambiornis exilis*	[34, 41]
18	*Marplesornis novaezealandiae*	[5.33, 15.97]
19	*Mesetaornis polaris*	[34, 41]
20	*Pachydyptes ponderosus*	[34.5, 36]
21	*Palaeeudyptes antarcticus*	[30.1, 34.5]
22	*Palaeeudyptes gunnari*	[34, 54]
23	*Palaeeudyptes klekowskii*	[34, 52.5]
24	*Palaeospheniscus bergi*	[9.7, 21]
25	*Palaeospheniscus biloculata*	[16, 21]
26	*Palaeospheniscus patagonicus*	[16, 21]
27	*Paraptenodytes antarcticus*	[21, 23]
28	*Perudyptes devriesi*	[38, 42]
29	*Platydyptes novaezealandiae*	[23, 26]
30	*Platydyptes marplesi*	[23, 30]
31	*Pygoscelis grandis*	[2.5, 8.6]
32	*Spheniscus megaramphus*	[6.3, 10]
33	*Spheniscus muizoni*	[9, 9.2]
34	*Spheniscus urbinai*	[5.6, 9.63]
35	*Waimanu manneringi*	[60.5, 61.6]
36	*Waimanu tuatahi*	[56, 60.5]

1.4 Validation

To validate the implementation of the models we simulated 100 trees under the FBD model with parameters:

$$t_{or} = 60 \quad \text{and} \quad (d, \nu, s, \rho) = (0.05, 0.6, 0.2, 0.8).$$

We discarded trees with less than five or more than 250 sampled nodes resulting in 96 trees. Then we simulated sequences (1000 sites) along the trees for extant samples only under the GTR model with parameters:

$$\pi_A = \pi_G = \pi_C = \pi_T = 0.25 \text{ and}$$

$$(\eta_{AG}, \eta_{AC}, \eta_{GT}, \eta_{AT}, \eta_{CG}, \eta_{CT}) = (1, 0.46, 0.45, 0.41, 1.85, 6.22)$$

and an uncorrelated relaxed log-normal model with parameters

$$\mu_{mol} = 4.36 \cdot 10^{-2} \text{ and}$$
$$\sigma_{mol} = 0.147$$

and morphological characters for all samples (200 characters per sample) assuming each character has ten states under a separate uncorrelated relaxed log-normal model with parameters

$$\mu_{morph} = 1.38 \cdot 10^{-2} \text{ and}$$
$$\sigma_{morph} = 0.728.$$

We chose parameter values that are close to the parameters estimated in the total-evidence analysis of the penguin dataset.

We analysed simulated morphological data matrix, molecular sequence data and fossil sampling dates to estimate phylogenies, FBD model parameters, clock and substitution model parameters. Nine analysis did not converge (one analysis with 239 sampled nodes and eight analyses with too few fossil or extant samples, i.e., fewer than four). The estimates for the remaining 87 converged analysis are summarised in Table 2.

2 Results

2.1 Divergence dates

We performed several different analyses under different model configurations and report the age of the MRCA of modern penguins in Table 3.

The estimated divergence dates and HPD intervals from the main analysis (same as Figure 4 in the main text) are given in Table 4. We also performed a total evidence analysis under the birth-death model without sampled ancestors and estimated slightly older divergence dates (see Figure 1).

Table 2: The results of the simulation study. The accuracy and precision is very poor for the morphological clock rate. After further removal of four analyses which required more time to converge and that have too few fossil or extant samples (fewer than five) this parameter estimate (numbers in brackets) became more accurate and precise.

parameter	true value	mean of relative errors	mean of relative biases	mean of relative 95% HPD lengths	95% HPD accuracy
t_{root}	51	0.05	$7.4 \cdot 10^{-3}$	0.22	93
t_{or}	60	0.13	$-4.35 \cdot 10^{-3}$	0.54	95
d	0.05	0.27	-0.21	1.57	100
ν	0.6	0.19	0.16	1.01	98
s	0.2	0.52	-0.42	2.4	95
ρ	0.8	0.11	0.08	0.66	100
μ_{mol}	$4.36 \cdot 10^{-2}$	0.09	-0.06	0.42	98
σ_{mol}	0.147	0.23	-0.03	1.38	97
μ_{morph}	$1.38 \cdot 10^{-2}$	13.69	-13.64	66.39	94
		(0.13)	(-0.09)	(0.62)	(94)
σ_{morph}	0.728	0.11	-0.04	0.59	99

Figure 1: Maximum clade credibility trees with common ancestor ages obtained after removing fossil lineages from the posterior trees (see section "Summarising trees" in the main text) and 95% HPD intervals for divergence times from total-evidence analyses under the birth-death model with sampled ancestors (blue phylogeny) and without (red phylogeny). The divergence ages estimated under the birth-death model without sampled ancestors are slightly older.

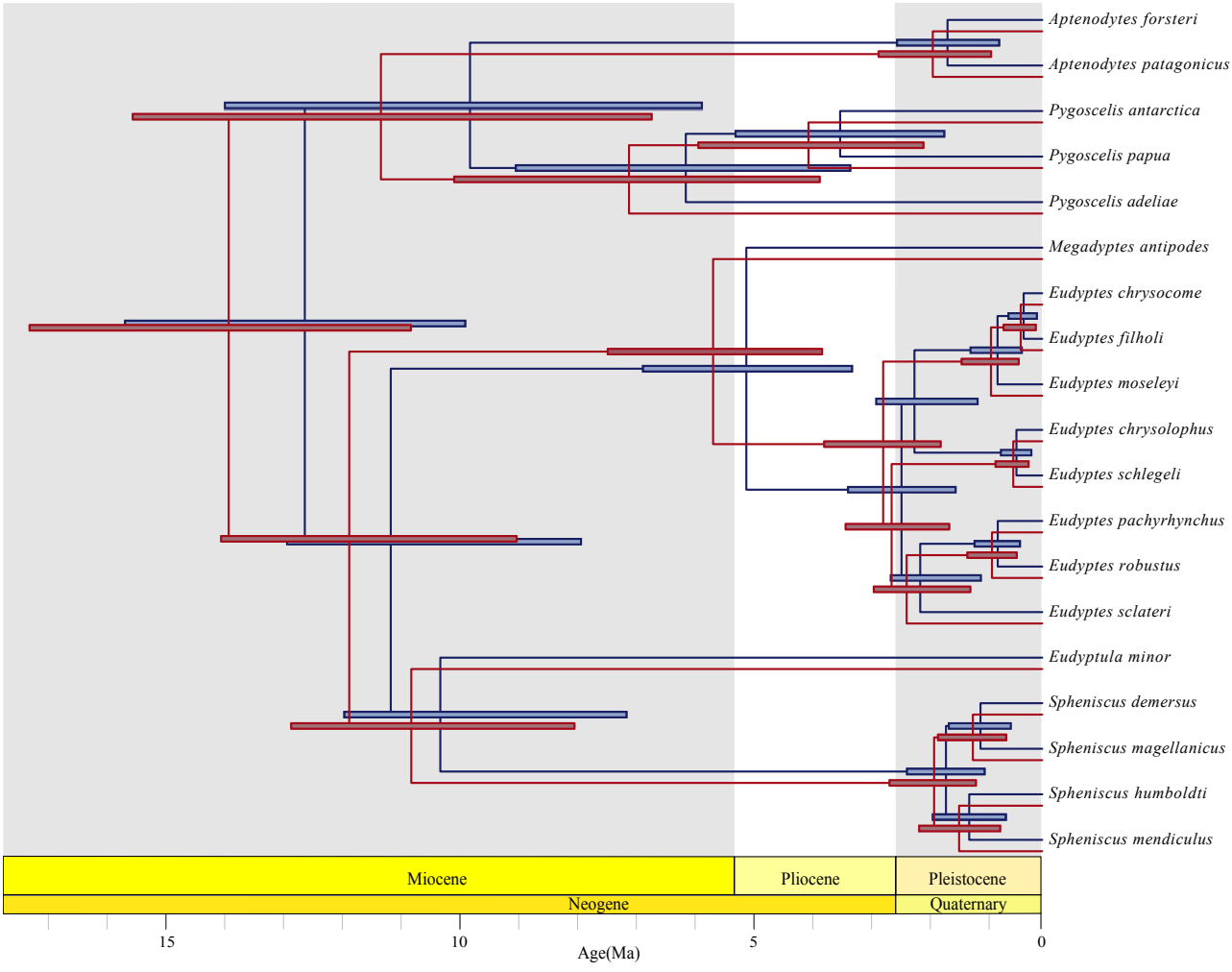


Table 3: The ages of the most recent common ancestor of extant penguins for eight analysis of morphological data, three analysis of combined data, and one analysis of combined data without stem fossil. The analysis numbers are as in Table 2 of the main text and the abbreviations are explained in the capture to Figure 6 in the main text.

Analysis #	Mean	95% HPD
1	13	[11.43, 14.49]
2	12.82	[11.35, 14.39]
3	12.45	[10.7, 14.36]
4	11.96	[10.06, 13.83]
5	12.47	[10.74, 14.48]
6	12.5	[10.75, 14.45]
7	11.91	[10.16, 13.76]
8	12.79	[10.48, 15.7]
8+dna R	12.66	[9.91, 15.75]
8BD+dna R	13.95	[10.83, 17.36]
8+dna S	13.46	[10.21, 16.56]
8+dna R crown only	22.84	[14.23, 33.56]

2.2 Sensitivity to the choice of prior distributions for the net diversification rate

Given enough comparative data of fossil and living samples, the FBD model allows us to estimate all hyper-parameters [20]. Thus, the true value of a parameter will be in the 95% HPD interval 95% of the time for analyses with uninformative priors for the FBD model hyper-parameters. Broad uninformative priors (for diversification rate which takes values from zero to infinity, for example) increase the time required for the convergence of an MCMC run. This is not an issue for the penguin dataset as it is relatively small. However, since we estimated marginal likelihoods for different models, the convergence time of a single analysis became important. To reduce this time we used a log-normal prior distribution for the net diversification rate. We investigated the impact of different prior distributions on the estimate of this parameter and on the parameter of the primary interest — the age of the MRCA of extant penguins. The results are shown in Figure 2 and Figure 3.

Figure 2: Prior and posterior probability densities for the net diversification rate for analysis 1 from Table 2 in the main text under four different prior distributions.

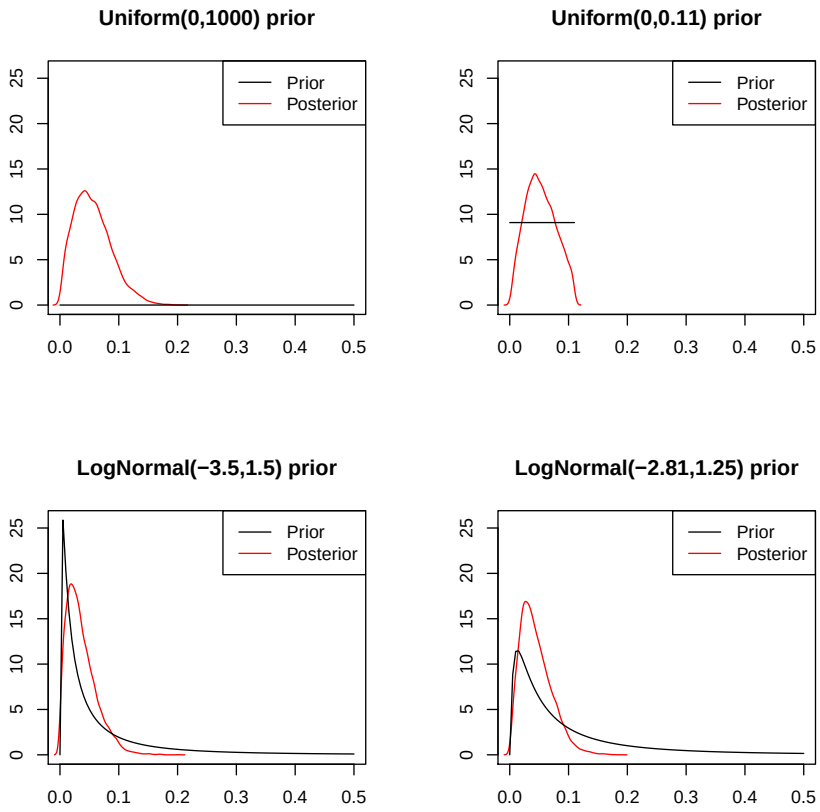


Figure 3: The estimates of the MRCA age of extant penguins for analysis 1 from Table 2 in the main text under four different prior distributions for the net diversification rate. The x-axis is labeled with the prior distributions used for the net diversification rate. These different prior distributions have little effect on the age of the MRCA.

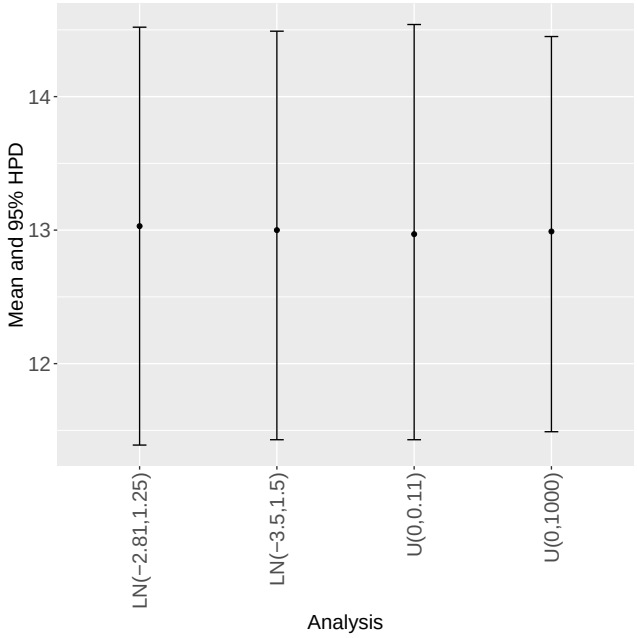


Table 4: Estimated divergence dates for extant penguin lineages inferred from the total-evidence analysis including all fossil and extant taxa. Node numbers correspond to the numbers in circles in Figure 4 of the main text.

Node	Mean	95% HPD
1	12.66	[9.91, 15.75]
2	11.19	[7.92, 12.97]
3	10.34	[7.14, 11.98]
4	9.83	[5.84, 14.04]
5	6.12	[3.3, 9.04]
6	5.08	[3.27, 6.86]
7	3.47	[1.68, 5.27]
8	2.42	[1.49, 3.33]
9	2.2	[1.11, 2.85]
10	2.1	[1.05, 2.6]
11	1.65	[0.99, 2.32]
12	1.63	[0.74, 2.49]
13	1.06	[0.63, 1.88]
14	1.06	[0.54, 1.6]
15	0.77	[0.35, 1.23]
16	0.76	[0.38, 1.16]
17	0.44	[0.19, 0.71]
18	0.32	[0.09, 0.58]

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